C PROGRAMMING MINI PROJECT REPORT

ATM Banking System

Project Information

Field	Details
Course Name	C Programming
Project Title	ATM Banking System
Project Type	Console-Based Application
Programming Language	C
Development Environment	Visual Studio Code / Code::Blocks / Dev C++
Version Control	Git & GitHub
Team Size	1–5 Students
Project Duration	1 Weeks
Difficulty Level	Beginner to Intermediate

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1. Problem Statement

Existing Problem

In many learning environments, students practice C programming by writing small programs that demonstrate only one concept at a time, such as loops, arrays, or functions. These individual programs help understand basic syntax but do not show how different C programming concepts can work together to solve a real-world problem. As a result, students often find it difficult to apply multiple concepts in a single project.

Proposed Solution

The proposed solution is to develop a Console-Based ATM Banking System using the C programming language. The system allows customers to log in securely using an account number and PIN to perform banking operations such as balance inquiry, deposit, withdrawal, money transfer, PIN change, and mini statement generation. Customer records are stored permanently using binary file handling, ensuring that data remains available even after the program is closed.

Target Users

- Students learning C Programming
- Faculty Members
- Lab Instructors
- Beginners who want to understand file handling and menu-driven programming

Expected Benefits

- Better understanding of practical C programming.
- Demonstrates the use of structures, arrays, functions, loops, and file handling.
- Provides permanent storage using binary files.
- Improves logical thinking and problem-solving skills.
- Gives experience in developing a real-world console application.

2. Project Objectives

Objective ID	Objective
OBJ-01	Create new customer accounts using the Bank Module.
OBJ-02	Store customer information securely in a binary file (bank.dat).
OBJ-03	Allow customers to log in using Account Number and PIN.
OBJ-04	Display customer account balance.

Objective ID	Objective
OBJ-05	Deposit money into the customer account.
OBJ-06	Withdraw money with balance validation.
OBJ-07	Transfer money between customer accounts.
OBJ-08	Allow customers to change their ATM PIN securely.
OBJ-09	Display a mini statement containing account details.
OBJ-10	Save and retrieve customer records using binary file handling.

3. User Stories

User Story ID	User Story	Acceptance Criteria	Priority
US-01	As a Bank Employee, I want to create a new customer account so that customers can use ATM services.	Customer account is created successfully with a unique account number.	High
US-02	As a Customer, I want to log in using my account number and PIN so that only I can access my account.	Login is successful only when valid credentials are entered.	High
US-03	As a Customer, I want to check my account balance so that I know my available funds.	Current balance is displayed correctly.	High
US-04	As a Customer, I want to deposit money into my account.	Deposit amount is added successfully and updated in the file.	High
US-05	As a Customer, I want to withdraw money from my account.	Withdrawal is allowed only when sufficient balance is available.	High
US-06	As a Customer, I want to transfer money to another account.	Money is transferred only if the receiver account exists and balance is sufficient.	High
US-07	As a Customer, I want to change my ATM PIN for security.	PIN changes successfully after verification of the old PIN.	Medium
US-08	As a Customer, I want my account data to remain saved after the program closes.	Customer data is stored and loaded successfully using binary file handling.	High

4. Functional Requirements

Requirement ID	Requirement	Status
FR-01	Create Customer Account	✓
FR-02	View Customer Accounts	✓
FR-03	Customer Login	✓
FR-04	Balance Inquiry	✓
FR-05	Deposit Money	✓
FR-06	Withdraw Money	✓
FR-07	Transfer Money	✓
FR-08	Change ATM PIN	✓
FR-09	Mini Statement	✓
FR-10	Save Customer Records to Binary File	✓
FR-11	Load Customer Records from Binary File	✓

5. Non-Functional Requirements

Category	Requirement
Performance	The application should respond quickly for all ATM operations and handle customer records efficiently.
Reliability	Customer information should be stored accurately without data loss.
Usability	The program should provide a simple menu-driven interface that is easy to use.
Maintainability	The code should be modular using separate functions for each operation.
Portability	The application should compile and execute on Windows and Linux systems supporting a C compiler.
Security	Login should require a valid account number and PIN. Input validation should prevent invalid data entry.

6. Project Modules

Module	Description	C Concepts Used
Account Setup Module	Creates new customer accounts, generates unique account numbers, accepts customer details, and stores them in the binary file.	Structures, Functions, File Handling
Login Module	Authenticates customers using account number and PIN before allowing access to ATM services.	Arrays, Loops, Conditional Statements
Balance Inquiry Module	Displays the customer's account number, name, and current account balance.	Functions
Deposit Module	Accepts a valid deposit amount, updates the account balance, and saves the changes.	Arithmetic Operators, File Handling
Withdrawal Module	Checks available balance before allowing cash withdrawal and updates the remaining balance.	Conditional Statements, Functions
Money Transfer Module	Transfers money from one customer account to another after validating the receiver's account and sufficient balance.	Arrays, Loops, Functions
PIN Change Module	Allows customers to change their ATM PIN after verifying the existing PIN.	Conditional Statements
Mini Statement Module	Displays a summary of the customer's account information.	Functions
File Management Module	Loads customer records from bank.dat during program startup and saves updated records after transactions.	Binary File Handling

7. C Programming Concepts Used

Concept	Implementation in Project
Variables	Used to store account number, PIN, balance, amount, menu choice, and other temporary values.
Data Types	int for account number and PIN, float for balance and transaction amounts, char for customer names.
Structures	struct Account stores all customer information together as a single record.
Arrays	An array of structures (accounts[MAX]) stores multiple customer records in memory.
Functions	Each banking operation such as login, deposit, withdrawal, transfer, and PIN change is implemented using separate functions.

Concept	Implementation in Project
Loops	while and for loops are used for menu repetition, searching customer accounts, and reading file records.
Switch Case	Used to display menu options and execute the selected banking operation.
Conditional Statements	Used for PIN validation, balance checking, login verification, and input validation.
Binary File Handling	fopen(), fread(), fwrite(), fclose(), and rewind() are used to permanently store customer data in bank.dat.

8. Expected Program Features

Feature	Description
Create Customer Account	Bank employee can register a new customer with account number, name, PIN, and opening balance.
Secure Login	Customers log in using account number and a valid 4-digit PIN.
Balance Inquiry	Displays the current account balance of the logged-in customer.
Deposit Money	Adds money to the customer's account after validating the amount.
Withdraw Money	Allows withdrawal only if sufficient balance is available.
Transfer Money	Transfers funds securely between two customer accounts.
Change ATM PIN	Customers can change their PIN after entering the correct current PIN.
Mini Statement	Displays account details and available balance.
Permanent Data Storage	Customer information remains saved even after closing the application.
Exit	Safely closes the application after saving updated records.

9. Test Cases

Test Case ID	Scenario	Input	Expected Output	Status
TC-01	Create New Account	Valid customer details	Account created successfully	Pass
TC-02	Login	Correct account number and PIN	Login successful	Pass
TC-03	Login	Incorrect PIN	Error message displayed	Pass
TC-04	Deposit Money	₹1000	Balance updated successfully	Pass

Test Case ID	Scenario	Input	Expected Output	Status
TC-05	Withdraw Money	Amount less than balance	Cash withdrawn successfully	Pass
TC-06	Withdraw Money	Amount greater than balance	"Insufficient Balance" message displayed	Pass
TC-07	Transfer Money	Valid receiver account	Transfer completed successfully	Pass
TC-08	Transfer Money	Invalid receiver account	"Receiver Account Not Found" message displayed	Pass
TC-09	Change PIN	Correct old PIN and valid new PIN	PIN updated successfully	Pass
TC-10	Save & Reload Data	Restart the application	Customer records loaded successfully	Pass

10. Expected Deliverables

Deliverable	Description
Source Code	Complete ATM Banking System written in C (account_setup.c and main.c).
Project Report	Detailed documentation in PDF/DOCX format.
PowerPoint Presentation	Project presentation explaining objectives, modules, workflow, and demonstration.
GitHub Repository	Public repository containing all project files and documentation.
README File	Complete project documentation with setup instructions and project features.
Output Screenshots	Screenshots of Account Creation, Login, ATM Menu, Deposit, Withdrawal, Transfer, and Mini Statement.
Test Case Report	Record of executed test cases with expected and actual outputs.

11. GitHub Repository Structure

```
ATM-Banking-System/
│
├── account_setup.c          # Bank Employee Module
├── main.c                   # ATM Module
├── bank.dat                 # Binary Customer Database
├── README.md                # Project Documentation
├── .gitignore               # Ignore executable files
│
├── Report/
│   └── ATM_Banking_System_Report.pdf
│
├── PPT/
│   └── ATM_Banking_System_Presentation.pptx
│
└── Output/
    ├── Account_setup.png
    ├── Login&Balance.png
    ├── ATM_Menu.png
    ├── Deposit&withdraw.png
    ├── TransferMoney&changePin.png
```

12. Challenges Faced During Development

During the development of the ATM Banking System, the following challenges were encountered:

- Understanding binary file handling and storing customer records permanently.
- Generating unique account numbers automatically.
- Implementing secure login using account number and PIN.
- Updating customer balances after every transaction.
- Managing multiple customer records using structures and arrays.
- Validating user inputs to prevent incorrect data entry.
- Dividing the project into separate Bank Module and ATM Module for better organization.

These challenges helped improve my understanding of C programming and problem-solving techniques.

13. Future Enhancements

Although the current project fulfills all the required objectives, the following improvements can be added in the future:

- Admin Login for bank employees.
- Transaction History for each customer.
- Account Search and Update features.
- Account Deletion facility.
- PIN Encryption for better security.

- Date and Time stamps for transactions.
- Receipt Generation after each transaction.
- Database integration using MySQL instead of binary files.
- Graphical User Interface (GUI) using C++ or Java.
- OTP-based authentication for secure login.

14. Expected Learning Outcomes

After completing this project, the following learning outcomes were achieved:

Outcome ID	Learning Outcome
LO-01	Applied C programming concepts to develop a practical application.
LO-02	Understood modular programming using functions.
LO-03	Learned how to use structures to organize related data.
LO-04	Gained experience with arrays for storing multiple records.
LO-05	Implemented binary file handling for permanent data storage.
LO-06	Improved logical thinking and debugging skills.
LO-07	Learned menu-driven programming using loops and switch-case statements.
LO-08	Understood authentication and input validation techniques.
LO-09	Practiced using Git and GitHub for version control.
LO-10	Improved confidence in explaining technical concepts during project presentations and viva examinations.

15. Conclusion

The ATM Banking System is a console-based application developed using the C programming language to demonstrate the practical implementation of fundamental programming concepts. The project successfully integrates structures, arrays, functions, loops, conditional statements, switch-case statements, and binary file handling into a single application.

The system allows customers to securely log in using their account number and PIN and perform essential banking operations such as balance inquiry, deposit, withdrawal, money transfer, PIN change, and viewing a mini statement. Customer information is stored permanently in a binary file, ensuring data persistence between program executions.

Developing this project improved my understanding of modular programming, file handling, data organization using structures, and logical problem-solving. It also provided practical experience in designing and implementing a menu-driven application using C.

Overall, this project fulfilled its objectives and served as an excellent learning experience for applying classroom concepts to a real-world scenario.

17. References

The following resources were used while developing this project:

- C Programming Lecture Notes provided by the Department.
- GCC Compiler Documentation.
- Git and GitHub Official Documentation.
- Microsoft Visual Studio Code Documentation.
- Personal coding practice and laboratory experiments.

18. Submission Checklist

Item	**Status**
Source Code Completed	☑
Code Compiles Successfully	☑
Test Cases Executed	☑
README Prepared	☑
GitHub Repository Created	☑
Minimum 10 Commits Completed	☑
Project Report Submitted	☑
PowerPoint Prepared	☑
Output Screenshots Added	☑
Viva Preparation Completed	☑